



The monkey on the Rope

Introduction

This is a beautifully simple puzzle, originally presented by Lewis Carroll, which is simple to describe but fiendishly difficult to solve. It challenges everything you think you know about the simple mechanics of ropes, weights and pulleys. Test your friends - they'll all think it's obvious, but they'll all disagree!

The puzzle is this:

A monkey hangs on a rope. The rope goes up over a pulley, and on the other end is a weight, which exactly balances the monkey. Everything is balanced, and is initially stationary. Then, the monkey tries to climb the rope. What happens?

Possible answers

At first this seems like a simple question, and to make it easier we'll first just look at the direction of movement, if any. Then we have the following possibilities:

- The weight moves up, it moves down, or it stays where it is
- The monkey moves up, it moves down, or it stays where it is

So here's your activity - think about the puzzle yourself, and come to a decision. Which direction do things move? Once you have an answer to that, you can think about how fast the things move - do they go faster and faster or stay at the same speed? What happens when the monkey stops trying to climb? Further questions then come up - where does the energy come from and where does it go? Is momentum conserved? Is angular momentum conserved?

The good thing about this puzzle is that the questions keep on presenting themselves. Does it matter how fast the monkey tries to climb? Is it possible for the monkey to control how fast the rope moves? And so on. Think up your own, until you're completely happy with your answer.

Pedants' corner

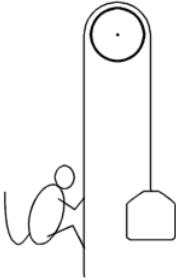
Inevitably we have to make some simplifying assumptions to make this problem easier. We want to consider the simplest, ideal conditions first, to see what the basic outcome is, and then modify this slightly if we need to later. The assumptions we make should be obvious, but in case of doubt here they are:

- The pulley is frictionless, so the rope is not slowed down or prevented from moving by the pulley.
- There is friction between the monkey and the rope, so the monkey can grip the rope without problems.
- The rope has no mass, so it doesn't matter how much rope is on one side of the pulley or the other.
- The rope is inextensible, so there is no bouncing and no hysteresis effects to consider.
- The monkey does not swing on the rope, but instead just tries to climb.
- The monkey's weight is constant, and not affected by exertion or time.
- The gravitational field is constant, so weights do not change with changing position.
- Relativistic effects can be ignored.

Also note that this is not a word game, don't look for a trick in the words used, just think about the problem described.

Solution

Once you have your answer (and not before!), see if you agree with the [solution](#). Also, please feel free to submit your thoughts on this problem via email to mail@activityworkshop.net.



A monkey, a rope, a pulley and a weight. Simple, eh?

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